

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
End-semester Examination
Numerical Methods (CS-106)
7th March 2023

Time: 3 hours

Total Marks: 50

PART - A

Answer all questions. ($5 \times 4 = 20$ marks)

Q1. How many positive roots does the equation $3x^3 - 2x^2 - x - 5 = 0$ have?

- (a) 2 (b) 3
(c) 1 (d) 0

Q2. The positive real roots of the equation $x^3 - 3x + 1 = 0$ lie in the intervals:

- (a) (0, 1) & (1, 2) (b) (2, 1) & (1, 3)
(c) (0, 2) & (1, 3) (d) (0, 1) & (2, 3)

Q3. Using Newton Raphson's method, the the smallest positive root of the equation $f(x) = x^3 - 5x + 1 = 0$ is:

- (a) 0.201640 (b) 0.120768
(c) 0.301640 (d) 0.102640

Q4. The solution of the system of equations

$$x_1 + 10x_2 - x_3 = 3$$

$$2x_1 + 3x_2 + 20x_3 = 7$$

$$10x_1 - x_2 + 2x_3 = 4$$

using the Gauss elimination with partial pivoting.

- (a) $x_1 = 0.53752, x_2 = 0.48940, x_3 = 0.36918$
 (b) $x_1 = 0.37512, x_2 = 0.28940, x_3 = 0.26908$
 (c) $x_1 = 0.17512, x_2 = 0.38194, x_3 = 0.32690$
 (d) $x_1 = 0.27432, x_2 = 0.18953, x_3 = 0.13469$

Q5. Using Lagrange's formula, the quadratic polynomial that takes the following values are:

$$x_0 = 0, y_0 = 0$$

$$x_1 = 1, y_1 = 1$$

$$x_2 = 3, y_2 = 0$$

(a) $6x^2 + 4x - 1$

(b) $4x^2 - 3x + 2$

(c) $\frac{-1}{2}(x^2 - 3x)$

(d) $\frac{1}{2}(3x^2 - x)$

Part-B

Answer all questions. (10 marks each, 10 = 30)

Q6. The following data gives the velocity of a particle for 20 seconds at an interval of 5 seconds. Find the initial acceleration using the entire data.

Time (sec)	0	5	10	15	20
Velocity (m/sec)	0	3	14	69	228

Q7. Find an approximation to $y(1.6)$, for the initial value problem

$$y' = x + y^2, y(1) = 1$$

using the Euler method with $h = 0.1$.

Q8. Derive the trapezium rule using the Lagrange linear interpolating polynomial. Find the approximate value of

$$I = \int_0^1 f(x) dx$$

where

$$f(x) = \frac{1}{1+x}$$

using the trapezium rule with 4 equal subintervals.

$$l_0(x) = \frac{x-b}{a-b}$$

$$l_1(x) = \frac{x-a}{b-a}$$

$$I = \int_a^b f(x) dx$$

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End-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
12th December 2024

Time: 3 hours

Total Marks: 50

Answer all questions. ($5 \times 10 = 50$ marks)

Q 1. Use Newton's forward difference formula for the given values to find $f(0.5)$ and $f(1.5)$

x	0	1	2	3	4
f(x)	1	7	23	55	109

Q 2. Find the solution of the system of equations

$$2x + y + 2z = 10$$

$$x + 2y + z = 8$$

$$3x + y - z = 2$$

using the Gauss-Jordan method.

Q 3. Find the interval in which the positive roots of the equation $x^2 + 5 = e^x$ lie. Find the smallest positive roots of the equation using Newton-Raphson method.

Q 4. Use the Runge-Kutta method to find approximate values of the solution of the given initial value problem at the points $x_i = x_0 + ih$, where x_0 is the point where the initial condition is imposed and $i = 1, 2$

$$y' = 2x^2 + 3y^2 - 2$$

$$y(2) = 1;$$

Assume the step length $h = 0.05$.

Q 5. Compute

$$I = \int_1^2 \frac{1}{x} dx$$

using trapezoidal and Simpson's rules with $n = 10$, $a = 1$, $b = 2$. Compare with exact solution.

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
CS - 406 Numerical Methods
End-semester Examination
5th December 2025

Time: 3 hours

Total Marks: 50

Answer all questions ($5 \times 10 = 50$ marks) .

Q 1. Find the positive roots of the equation $x^3 - 3x + 1 = 0$ using Newton Raphson method.

Q 2. Find the solution of the system of equations

$$\begin{aligned}x_1 + x_2 - x_3 &= 2 \\2x_1 + 3x_2 + 5x_3 &= -3 \\3x_1 + 2x_2 - 3x_3 &= 6\end{aligned}$$

using the Gauss-Jordan method.

Q 3. Using Lagrange's formula, find the interpolating polynomial that takes the following values.

$$x_0 = 0, y_0 = 0; x_1 = 1, y_1 = 1; x_2 = 3, y_2 = 0$$

Q 4. Derive the trapezium rule using the Lagrange linear interpolating polynomial. Find the approximate value of

$$I = \int_0^1 \frac{x}{x^3 + 10} dx$$

using the composite trapezoidal rule with 2 equal subintervals.

Q 5. Find the solution at $x = 0.4$ of the initial value problem using the fourth order Runge-Kutta method.

$$\frac{dy}{dx} = -2xy^2, y(0) = 1$$

Assume the step length $h = 0.2$.